

BACKLOG BUDDY ? STUDY MATERIAL

Digital Signal Processing ? Unit-wise Lecture Notes

Subject Code: EC501 Semester: 5

.....
Condensed notes covering all five units: Discrete Time Signals and Systems, DFT and FFT Algorithms, FIR Filter Design, IIR Filter Design, Digital Signal Processors.

Unit 1: Discrete Time Signals and Systems

- Definition, key concepts and significance of Discrete Time Signals and Systems
- Important diagrams / derivations / algorithms for Discrete Time Signals and Systems
- Solved examples and previous-year questions on Discrete Time Signals and Systems

Unit 2: DFT and FFT Algorithms

- Definition, key concepts and significance of DFT and FFT Algorithms
- Important diagrams / derivations / algorithms for DFT and FFT Algorithms
- Solved examples and previous-year questions on DFT and FFT Algorithms

Unit 3: FIR Filter Design

- Definition, key concepts and significance of FIR Filter Design
- Important diagrams / derivations / algorithms for FIR Filter Design
- Solved examples and previous-year questions on FIR Filter Design

Unit 4: IIR Filter Design

- Definition, key concepts and significance of IIR Filter Design
- Important diagrams / derivations / algorithms for IIR Filter Design
- Solved examples and previous-year questions on IIR Filter Design

Unit 5: Digital Signal Processors

- Definition, key concepts and significance of Digital Signal Processors
- Important diagrams / derivations / algorithms for Digital Signal Processors
- Solved examples and previous-year questions on Digital Signal Processors

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.